The Convexity Theorem for the Exponential.

From Weighted AM–GM and Midpoint Identities to the Lean's Slope Argument

Brahimi Mahdi Tahar

University of Mohamed Boudiaf of M'sila, Algeria

mahditahar.brahimi@univ-msila.dz

Abstract

We collect ten routes to the convexity theorem for the exponential function

$$\exp : \mathbb{R} \to \mathbb{R}_{>0}.$$

- Three are constructive in Bishop's sense:
  The weighted AM–GM route, a direct term-by-term power-series proof on the non-negative reals followed by a translation, and a midpoint convexity proof via the algebraic identity
  $$\forall x \in \mathbb{R},\ \forall y \in \mathbb{R} : \quad (e^{x/2} - e^{y/2})^2 \geq 0.$$
- Three are classical differential proofs:
  The second-derivative test, the monotonicity of the first derivative, and the supporting-line inequality
  $$\forall x \in \mathbb{R},\ \forall y \in \mathbb{R} : \quad e^x \geq e^y + e^y(x - y).$$
- Three are further routes:
  The limit definition with Bernoulli's inequality, the inverse of the concave logarithm, and a pointer to the Lean's source.
- The tenth is a detailed mathematical adaptation of the Lean's proof, using the slope criterion and the strict supporting-line inequality
  $$\forall t \in \mathbb{R} : \quad t \neq 0 \Rightarrow e^t > 1 + t.$$

The power-series definition is used throughout:

$$\exp(x) = L \iff \forall \varepsilon \in \mathbb{R}_{>0},\ \exists N \in \mathbb{N},\ \forall m \in \mathbb{N} :\ m \geq N \Rightarrow \left| L - \sum_{k=0}^{m} \frac{x^k}{k!} \right| < \varepsilon.$$

The theorem is RCA_0-provable under Bishop–Cauchy coding.
The classical differential routes (4 and 5), which invoke the second-derivative test and the mean value/FTC machinery, are WKL_0-provable.
Route 6, although classical in style, stays within the power series and is RCA_0-provable.

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Contents

1. Introduction 1

2. Constructive proofs 3

2.1. Weighted AM–GM (First proof) 3

2.2. Direct series on the non-negative reals (Second proof) 5

2.3. Midpoint convexity via an algebraic square (Third proof) 7

3. Classical differential proofs 9

3.1. Second-derivative test (Fourth proof) 9

3.2. The derivative is strictly increasing (Fifth proof) 10

3.3. Supporting-line tangent inequality (Sixth proof) 10

4. Further proofs 11

4.1. Limit definition and Bernoulli's inequality (Seventh proof) 11

4.2. Inverse of the concave logarithm (Eighth proof) 12

4.3. Pointer to the Lean's proof (Ninth proof) 12

5. The Lean's proof 13

5.1. Source code. 13

5.2. Auxiliary lemma (strict supporting-line inequality) 13

5.3. Auxiliary lemma (slope criterion) 14

5.4. Proof of the exp convexity (Lean slope argument) 15

6. Reverse-mathematical classification proofs 17

6.1. Coding conventions 18

6.2. Field and order axioms 19

6.3. Postulates. 20

6.4. Definitions 22

6.5. Statements (the ten routes re-classified) 22

6.6. Proofs (dependency tables) 23

1. Introduction

The convexity of the exponential function is a classical fact, usually derived from the strict positivity of its second derivative.
In Bishop's constructive mathematics that argument is unavailable, because the second-derivative test presupposes a constructive theory of differentiation and the mean value theorem, both of which require additional hypotheses in Bishop's framework.
The purpose of this collection is to record ten routes to the same conclusion:
Each independent and each verifying the convexity inequality by different means.
The geometric content of the inequality is shown in Figure 1.
The Cauchy-Bishop power-series characterisation of the exponential function is:

$$\forall x \in \mathbb{R},\ \forall \varepsilon \in \mathbb{R}_{>0},\ \exists N \in \mathbb{N},\ \forall m \in \mathbb{N} : \quad m \geq N \Rightarrow \left|\exp(x) - \sum_{k=0}^{m} \frac{x^k}{k!}\right| < \varepsilon, \tag{1.1}$$

and is used throughout this document.
We rely on the standard consequences of absolute convergence on bounded intervals:

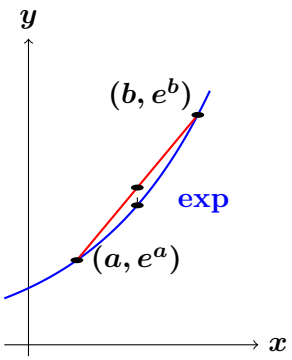


Figure 1: Convexity of exp: the chord joining (a, e^a) and (b, e^b) lies above the graph, in particular above the midpoint.

(1) Positivity:

$$\forall x \in \mathbb{R} : \ \exp(x) > 0. \tag{1.2}$$

(2) Multiplicativity:

$$\forall x \in \mathbb{R}, \forall y \in \mathbb{R} : \ \exp(x + y) = \exp(x) \exp(y). \tag{1.3}$$

(3) The identity:

$$\forall x \in \mathbb{R}, \forall t \in [0, 1] : \ \exp(tx) = (\exp x)^t. \tag{1.4}$$

(4) Uniform convergence on every bounded interval:

For $R \in \mathbb{R}_{>0}$, define a finite real

$$M_n(R) := \sup_{x \in [-R,R]} \left| \exp(x) - \sum_{k=0}^{n} \frac{x^k}{k!} \right| \qquad (n \in \mathbb{N}).$$

The function under the supremum is continuous on $[-R, R]$, so $M_n(R)$ is a finite real.

Then:

$$\forall R \in \mathbb{R}_{>0},\ \forall \varepsilon \in \mathbb{R}_{>0},\ \exists N \in \mathbb{N},\ \forall n \in \mathbb{N} : \quad n \geq N \Rightarrow M_n(R) < \varepsilon.$$

A finite modulus is supplied by the tail bound

$$\forall R \in \mathbb{R}_{>0},\ \forall x \in [-R, R],\ \forall n \in \mathbb{N} : \quad n+2 > R \Rightarrow \left| \exp(x) - \sum_{k=0}^{n} \frac{x^k}{k!} \right| \leq \frac{R^{n+1}}{(n+1)!} \left(1 - \frac{R}{n+2}\right)^{-1}.$$

Indeed, if N is chosen so that for every $n \geq N$

$$\frac{R^{n+1}}{(n+1)!} < \frac{\varepsilon}{2} \quad \text{and} \quad \left(1 - \frac{R}{n+2}\right)^{-1} < 2,$$

then $M_n(R) < \varepsilon$ for every $n \geq N$.

These are established in the constructive setting in Bishop–Bridges [1] and elsewhere.

Constructive inequalities of means are treated in [6].

The ten proofs are organised as follows:

- Section 2 collects the three constructive proofs (AM–GM, direct series plus translation, midpoint convexity).
- Section 3 collects the three classical differential proofs (second derivative, first derivative monotone, supporting line).
- Section 4 collects three further proofs (limit definition with Bernoulli, inverse of the concave logarithm, and the pointer to Lean).
- Section 5 gives the detailed adaptation of the Lean's proof.
- Section 6 summarises the ten routes in tabular form and re-classifies each in second-order arithmetic.

Definition 1.1.

For $n \in \mathbb{N}$ and $x \in \mathbb{R}$, set

$$S_n(x) := \sum_{k=0}^{n} \frac{x^k}{k!}. \tag{1.5}$$

The exponential $\exp x$ is the unique real L satisfying

$$\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N : |L - S_n(x)| < \varepsilon. \tag{1.6}$$

For $R \geq 0$, set $T_n(R) := \sum_{k=n+1}^{\infty} R^k/k!$.

Definition 1.2 (Convexity and strict convexity).

Let $I \subseteq \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$. Define

$$\mathrm{Conv}(f, I) \iff \forall x, y \in I\ \forall t \in [0,1] : f(tx + (1-t)y) \leq t f(x) + (1-t) f(y). \tag{1.7}$$

Define
$\mathrm{StrictConv}(f, I) \iff \mathrm{Conv}(f, I) \wedge$

$$\forall x, y \in I\ \forall t \in (0,1):\ x \neq y \implies f(tx + (1-t)y) < tf(x) + (1-t)f(y).$$

For the constructive strict version, replace $x \neq y$ by $|x - y| > 0$.

Definition 1.3 (Compact-uniform continuity modulus).
For a compact interval $J \subseteq I$ and $f : I \to \mathbb{R}$, a modulus is a function $\mu_J : \mathbb{N} \to \mathbb{N}$ such that

$$\forall k \in \mathbb{N}\ \forall x, y \in J:\quad |x - y| < 2^{-\mu_J(k)} \implies |f(x) - f(y)| < 2^{-k}. \tag{1.8}$$

Definition 1.4 (Derivative).
Let $I \subseteq \mathbb{R}$ be open and $x \in I$. Define $f'(x) = L$ by

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall h \in \mathbb{R}:\quad \big(0 < |h| < \delta\ \wedge\ x + h \in I\big) \implies \left|\frac{f(x+h) - f(x)}{h} - L\right| < \varepsilon. \tag{1.9}$$

Definition 1.5 (C^k on an open interval).
For an open interval I, define $f \in C^k(I)$ if there exist continuous functions $g_0, \ldots, g_k : I \to \mathbb{R}$ such that

$$g_0 = f, \qquad g_j'(x) = g_{j+1}(x) \quad (x \in I,\ 0 \le j < k). \tag{1.10}$$

Definition 1.6 (Cauchy codes and base theories).
A fast Cauchy code for a real is a sequence $q : \mathbb{N} \to \mathbb{Q}$ with

$$|q_m - q_n| \le 2^{-n} \qquad (m \ge n). \tag{1.11}$$

Two such codes represent the same real if their difference tends to zero. On a rational compact interval, a modulated continuous-function code consists of coherent rational value approximations on rational inputs and a modulus as in Definition 4; evaluation at real inputs is by Cauchy approximation. On open domains use compatible codes on a compact exhaustion. Set

$$\mathsf{RCA}_0 := \text{basic second-order arithmetic} + \Delta^0_1\text{-comprehension} + \Sigma^0_1\text{-induction},$$
$$\mathsf{WKL}_0 := \mathsf{RCA}_0 + \mathsf{WKL},$$

where WKL asserts that every infinite binary tree has an infinite path. Real quantifiers in coded statements range over valid Cauchy codes.

2. Constructive proofs

2.1. Weighted AM–GM (First proof)

Lemma 2.1 (Weighted AM–GM, two variables).
For every $a \in \mathbb{R}_{>0}$, every $b \in \mathbb{R}_{>0}$, and every $t \in [0,1]$,

$$a^t\, b^{1-t} \ \le\ t\,a + (1-t)\,b, \tag{2.1}$$

where a^t and b^{1-t} denote the real powers `Real.rpow` $a\ t$ and `Real.rpow` $b\ (1-t)$.

Proof.

Rational case:

Suppose $t = m/n$ with $m, n \in \mathbb{N}$, $0 \le m \le n$, $n \ge 1$.

Applying the ordinary AM–GM inequality (see [3]) to the n positive numbers consisting of m copies of a and $n-m$ copies of b gives

$$\left(a^m b^{n-m}\right)^{1/n} \le \frac{ma + (n-m)b}{n}.$$

The left-hand side equals

$$a^{m/n} b^{(n-m)/n} = a^t b^{1-t},$$

and the right-hand side equals

$$\frac{m}{n}a + \frac{n-m}{n}b = ta + (1-t)b.$$

Hence ineq 2.1 holds whenever t is rational.

Real case:

Fix $t \in [0,1]$ and $\varepsilon > 0$.

The maps $s \mapsto a^s b^{1-s}$ and $s \mapsto sa + (1-s)b$ are continuous on $[0,1]$, so there is $\delta > 0$ such that, for every $t' \in [0,1] \cap \mathbb{Q}$ with $|t - t'| < \delta$,

$$\left|a^t b^{1-t} - a^{t'} b^{1-t'}\right| < \tfrac{\varepsilon}{2}, \qquad \left|(ta + (1-t)b) - (t'a + (1-t')b)\right| < \tfrac{\varepsilon}{2}.$$

Choose such a t'. By the rational case, $a^{t'} b^{1-t'} \le t'a + (1-t')b$.

Combining,

$$a^t b^{1-t} < a^{t'} b^{1-t'} + \tfrac{\varepsilon}{2} \le t'a + (1-t')b + \tfrac{\varepsilon}{2} < ta + (1-t)b + \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, ineq 2.1 follows.

□

Theorem 2.1 (Convexity via AM–GM).
For every $x \in \mathbb{R}$, every $y \in \mathbb{R}$, and every $t \in [0,1]$,

$$\exp\bigl(tx + (1-t)y\bigr) \;\le\; t\,\exp x + (1-t)\,\exp y. \tag{2.2}$$

Proof.

By multiplicativity and the identity $\exp(tx) = (\exp x)^t$ (see `Pos:ExpPow` in §6.3; for real t it follows from the rational case together with continuity of both $t \mapsto \exp(tx)$ and $t \mapsto (\exp x)^t$), we get

$$\exp\bigl(tx + (1-t)y\bigr) = (\exp x)^t (\exp y)^{1-t}.$$

Apply Lemma 2.1 with $a = \exp x > 0$ and $b = \exp y > 0$.

Then:

$$(\exp x)^t (\exp y)^{1-t} \le t \exp x + (1-t)\exp y,$$

which combined with the display gives ineq 2.2.

□

2.2. Direct series on the non-negative reals (Second proof)

Lemma 2.2 (Monomial convexity on the non-negative reals).

$$\forall n \in \mathbb{N}_{\geq 1},\ x, y \in \mathbb{R}_{\geq 0},\ \forall t \in [0,1]: \quad (tx + (1-t)y)^n \ \leq\ t\,x^n + (1-t)\,y^n. \tag{2.3}$$

Proof.

Base case:

For $n = 1$ both sides equal $tx + (1-t)y$, so ineq 2.3 holds with equality.

Inductive step:

Assume ineq 2.3 for some $n \geq 1$, and put

$$z := tx + (1-t)y.$$

Since $t \in [0,1]$ and $x, y \geq 0$, we have $z \geq 0$.

Hence

$$z^{n+1} = zz^n \leq z\big(tx^n + (1-t)y^n\big).$$

Comparing the right-hand side with the target bound

$$tx^{n+1} + (1-t)y^{n+1},$$

the gap is

$$tx^{n+1}+(1-t)y^{n+1}-z\big(tx^n+(1-t)y^n\big) = t(1-t)\big(x^{n+1}+y^{n+1}-xy^n-x^ny\big) = t(1-t)(x-y)(x^n-y^n).$$

Since $t(1-t) \geq 0$, and since $x, y \geq 0$ forces the two factors $x - y$ and $x^n - y^n$ to have the same sign, their product is ≥ 0.

Therefore

$$z\big(tx^n + (1-t)y^n\big) \leq tx^{n+1} + (1-t)y^{n+1}.$$

Combining with the inductive inequality gives

$$z^{n+1} \leq tx^{n+1} + (1-t)y^{n+1},$$

which is ineq 2.3 with n replaced by $n+1$.

□

Theorem 2.2 (Convexity on the non-negative reals from the series).
For $N \in \mathbb{N}$ and $u \in \mathbb{R}$, put

$$S_N(u) := \sum_{k=0}^{N} u^k/k!.$$

Then:

(1) Polynomial level :

$$\forall N \in \mathbb{N}, \forall x \in \mathbb{R}_{\geq 0}, \forall y \in \mathbb{R}_{\geq 0}, \forall t \in [0,1]: \ S_N(tx + (1-t)y) \ \leq\ t\,S_N(x) + (1-t)\,S_N(y). \tag{2.4}$$

(2) Exponential level :

$$\forall x \in \mathbb{R}_{\geq 0}, \forall y \in \mathbb{R}_{\geq 0}, \forall t \in [0,1] : \quad \exp(tx + (1-t)y) \;\leq\; t\,\exp x + (1-t)\,\exp y. \quad (2.5)$$

Proof.

Put

$$P_t(x,y) := tx + (1-t)y.$$

Since $x, y \geq 0$ and $t \in [0,1]$, we have $P_t(x,y) \geq 0$.

Proof of ineq 2.4

Fix $N \in \mathbb{N}$, $x, y \in \mathbb{R}_{\geq 0}$, $t \in [0,1]$.

By definition,

$$S_N(P_t(x,y)) = \sum_{k=0}^{N} \frac{P_t(x,y)^k}{k!}, \qquad t\,S_N(x) + (1-t)\,S_N(y) = \sum_{k=0}^{N} \frac{t\,x^k + (1-t)\,y^k}{k!}.$$

For every $k \in \mathbb{N}$ with $k \geq 1$, Lemma 2.2 gives

$$P_t(x,y)^k \leq t\,x^k + (1-t)\,y^k.$$

For $k = 0$ both sides equal 1.

Dividing by $k!$ and summing over $k = 0, \ldots, N$,

$$S_N(P_t(x,y)) \leq t\,S_N(x) + (1-t)\,S_N(y),$$

which is ineq 2.4.

Proof of ineq 2.5

Fix $x, y \in \mathbb{R}_{\geq 0}$, $t \in [0,1]$, and $\varepsilon > 0$.

By the ε-N characterisation of exp stated in the introduction, there are $N_1, N_2, N_3 \in \mathbb{N}$ such that, for every $N \in \mathbb{N}$,

$$\begin{aligned} N \geq N_1 &\implies \left|\exp(P_t(x,y)) - S_N(P_t(x,y))\right| < \tfrac{\varepsilon}{3}, \\ N \geq N_2 &\implies \left|\exp(x) - S_N(x)\right| < \tfrac{\varepsilon}{3}, \\ N \geq N_3 &\implies \left|\exp(y) - S_N(y)\right| < \tfrac{\varepsilon}{3}. \end{aligned}$$

Set

$$N \geq N_0, \quad N_0 := \max\{N_1, N_2, N_3\}.$$

Since $t \in [0,1]$,

$$\left|t\exp x - t\,S_N(x)\right| \leq |\exp x - S_N(x)| < \tfrac{\varepsilon}{3}, \qquad \left|(1-t)\exp y - (1-t)\,S_N(y)\right| < \tfrac{\varepsilon}{3}.$$

Combining these bounds with ineq 2.4,

$$\exp(P_t(x,y)) < S_N(P_t(x,y)) + \tfrac{\varepsilon}{3} \leq t\,S_N(x) + (1-t)\,S_N(y) + \tfrac{\varepsilon}{3} < t\exp x + (1-t)\exp y + \varepsilon.$$

As $\varepsilon > 0$ was arbitrary, ineq 2.5 follows.

□

Theorem 2.3 (Convexity on $\mathbb{R}$ via translation).
The inequality ineq 2.2 holds on all of $\mathbb{R}$ by translation.

Proof.

Fix $x, y \in \mathbb{R}$. By Archimedes' axiom there are $M_x, M_y \in \mathbb{N}$ with $|x| \le M_x$ and $|y| \le M_y$; put $M := \max\{M_x, M_y\}$. Then

$$x' := x + M \ge 0, \qquad y' := y + M \ge 0.$$

Since

$$tx + (1-t)y = tx' + (1-t)y' - M,$$

multiplicativity gives

$$\exp\big(tx + (1-t)y\big) = \exp\big(tx' + (1-t)y'\big)\exp(-M).$$

Apply Theorem 2.2 to $x', y' \ge 0$, multiply by $\exp(-M) > 0$, and use multiplicativity once more.

□

2.3. Midpoint convexity via an algebraic square (Third proof)

Lemma 2.3 (Midpoint convexity).
For every $x \in \mathbb{R}$ and every $y \in \mathbb{R}$,

$$\exp\left(\frac{x+y}{2}\right) \le \frac{\exp x + \exp y}{2}. \tag{2.6}$$

Proof.

The algebraic identity

$$\forall x \in \mathbb{R}, \forall y \in \mathbb{R}: \ \left(e^{x/2} - e^{y/2}\right)^2 \ge 0,$$

expands to

$$\forall x \in \mathbb{R}, \forall y \in \mathbb{R}: \ e^x + e^y \ge 2e^{(x+y)/2}.$$

Dividing by 2 gives ineq 2.6.

□

Theorem 2.4 (Convexity via midpoint convexity).
The inequality ineq 2.2 holds by midpoint convexity.

Proof.

Fix $x, y \in \mathbb{R}$.

For $t \in [0,1]$ put

$$P(t) := tx + (1-t)y.$$

The inequality of ineq 2.2 is

$$\exp(P(t)) \le t\exp x + (1-t)\exp y.$$

Step 1: dyadic t.

We prove: for every $n \in \mathbb{N}$ and every $k \in \{0, 1, \dots, 2^n\}$,

$$\exp\big(P(k/2^n)\big) \le \frac{k}{2^n}\exp x + \left(1 - \frac{k}{2^n}\right)\exp y. \tag{2.7}$$

Base case $n = 0$:

Then $k \in \{0,1\}$, and

$$\exp(P(0)) = \exp y, \qquad \exp(P(1)) = \exp x,$$

so ineq 2.7 holds with equality.

Inductive step:

Assume ineq 2.7 for some n, and let $k \in \{0,1,\dots,2^{n+1}\}$.

Case $k = 2j$ even:

Then

$$k/2^{n+1} = j/2^n$$

with

$$j \in \{0,\dots,2^n\},$$

so ineq 2.7 at $(n+1,k)$ is exactly the induction hypothesis at (n,j).

Case $k = 2j+1$ odd:

Then

$$j \in \{0,\dots,2^n - 1\}.$$

Set

$$u := \frac{j}{2^n}, \qquad v := \frac{j+1}{2^n}, \qquad \frac{u+v}{2} = \frac{2j+1}{2^{n+1}} = \frac{k}{2^{n+1}}.$$

Since P is affine,

$$P\big(\tfrac{u+v}{2}\big) = \tfrac{1}{2}\big(P(u) + P(v)\big).$$

The midpoint inequality ineq 2.6 with the two reals $P(u)$ and $P(v)$ gives

$$\exp\big(P(k/2^{n+1})\big) = \exp\Big(\tfrac{P(u)+P(v)}{2}\Big) \le \frac{\exp(P(u)) + \exp(P(v))}{2}.$$

Applying the induction hypothesis at (n,j) and $(n,j+1)$,

$$\begin{aligned}\frac{\exp(P(u)) + \exp(P(v))}{2} &\le \frac{\big(u\exp x + (1-u)\exp y\big) + \big(v\exp x + (1-v)\exp y\big)}{2} \\ &= \frac{u+v}{2}\exp x + \Big(1 - \frac{u+v}{2}\Big)\exp y,\end{aligned}$$

which is ineq 2.7 at $(n+1,k)$.

This closes the induction.

Step 2: arbitrary $t \in [0,1]$.

If $x = y$, then $P(t) = x$ and both sides of ineq 2.2 equal $\exp x$.

Assume $x \neq y$.

Fix $t \in [0,1]$ and $\varepsilon > 0$.

Continuity of exp at $P(t)$:

Since exp is continuous (uniform convergence of its series on bounded intervals), there exists $\eta > 0$ such that

$$|Q - P(t)| < \eta \implies |\exp(Q) - \exp(P(t))| < \tfrac{\varepsilon}{2}.$$

Set

$$\rho := \min\Big(\frac{\eta}{|x-y|},\ \frac{\varepsilon}{2\,|\exp x - \exp y|}\Big) \quad \text{(the second term is omitted if } \exp x = \exp y\text{)}.$$

By density of the dyadics in $[0,1]$, choose

$$u \in \{k/2^n : n \in \mathbb{N},\ 0 \le k \le 2^n\}$$

with $|u-t| < \rho$.

Two bounds:

On the one hand,

$$|P(u) - P(t)| = |u-t|\,|x-y| < \rho\,|x-y| \le \eta,$$

so

$$|\exp(P(u)) - \exp(P(t))| < \varepsilon/2.$$

On the other hand,

$$\big|(u\exp x + (1-u)\exp y) - (t\exp x + (1-t)\exp y)\big| = |u-t|\,|\exp x - \exp y| < \rho\,|\exp x - \exp y| \le \tfrac{\varepsilon}{2}.$$

Conclusion:

Combining these bounds with ineq 2.7 at u,

$$\exp(P(t)) < \exp(P(u)) + \tfrac{\varepsilon}{2} \le u\exp x + (1-u)\exp y + \tfrac{\varepsilon}{2} < t\exp x + (1-t)\exp y + \varepsilon.$$

As $\varepsilon > 0$ was arbitrary,

$$\exp(P(t)) \le t\exp x + (1-t)\exp y,$$

which is ineq 2.2.

□

3. Classical differential proofs

The proofs of this section are standard (see [4]). Routes 4 and 5 are not constructive in Bishop's sense because they invoke the mean value theorem or the second-derivative test, and they are WKL_0-provable. Route 6, although stated in the classical idiom of supporting lines, relies only on the power series and is RCA_0-provable.

3.1. Second-derivative test (Fourth proof)

Theorem 3.1.

exp is strictly convex on $\mathbb{R}$ by the second-derivative test.

Proof.

Since

$$\forall x \in \mathbb{R}:\ \ \exp''(x) = \exp(x) > 0,$$

a twice differentiable function with positive second derivative is strictly convex.

□

3.2. The derivative is strictly increasing (Fifth proof)

Theorem 3.2.

exp is strictly convex on $\mathbb{R}$ via a strictly increasing derivative.

Proof.

Since ineq 3.1 holds,

$$\forall x \in \mathbb{R}: \quad \exp'(x) = \exp(x) > 0, \tag{3.1}$$

the exponential function exp is strictly increasing. Since $\exp' = \exp$, the derivative $\exp'$ is itself strictly increasing.

A differentiable function on an interval is convex if and only if its derivative is increasing.

□

3.3. Supporting-line tangent inequality (Sixth proof)

Lemma 3.1.

For every $z \in \mathbb{R}$,

$$e^z \geq 1 + z. \tag{3.2}$$

Classical proof via the series.

If $z \geq 0$, then for every $N \in \mathbb{N}$,

$$\sum_{k=0}^{N} z^k/k! \geq 1 + z,$$

and the inequality passes to $\exp(z)$.

If $-1 \leq z < 0$, pair consecutive terms: for every $k \geq 1$,

$$\frac{z^{2k}}{(2k)!} + \frac{z^{2k+1}}{(2k+1)!} = \frac{z^{2k}}{(2k)!}\left(1 + \frac{z}{2k+1}\right) \geq 0,$$

because $z^{2k} \geq 0$ and $|z| \leq 1 \leq 2k+1$, so $1 + \frac{z}{2k+1} \geq 0$. Each consecutive pair is therefore non-negative, and the partial sums stay $\geq 1 + z$.

If $z < -1$, then $1 + z < 0 < e^z$ (by Pos:23), so the inequality holds trivially.

□

Theorem 3.3.

exp is convex on $\mathbb{R}$ via the supporting-line tangent inequality.

Proof.

A differentiable function f is convex if and only if

$$\forall x \in \mathbb{R}, \forall y \in \mathbb{R}: \quad f(x) \geq f(y) + f'(y)(x - y).$$

For $f = \exp,$ this becomes

$$\forall x \in \mathbb{R}, \forall y \in \mathbb{R}: \quad e^x \geq e^y + e^y(x - y).$$

Putting $z = x - y$ and dividing by $e^y > 0$ reduces this to ineq 3.2.

The geometric content of the criterion is shown in Figure 2.

□

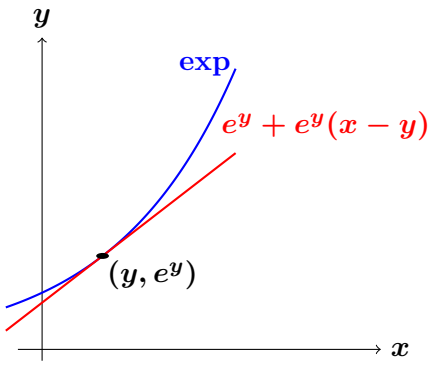


Figure 2: Supporting line: the tangent to exp at (y, e^y) lies below the graph.

4. Further proofs

4.1. Limit definition and Bernoulli's inequality (Seventh proof)

Theorem 4.1 (Convexity via the limit definition and Bernoulli's inequality).
exp is convex on $\mathbb{R}$.

Proof.

Step 1: limit definition of exp.

The exponential is the limit of the sequence $\left((1+x/n)^n\right)_n$, in the ε-N sense

$$\forall x \in \mathbb{R},\ \forall \varepsilon \in \mathbb{R}_{>0},\ \exists N \in \mathbb{N},\ \forall n \in \mathbb{N}: \quad n \geq N \implies \left|\left(1+\frac{x}{n}\right)^n - \exp(x)\right| < \varepsilon. \tag{4.1}$$

Step 2: the inequality $e^x \geq 1+x$.

Fix $x \in \mathbb{R}_{\geq -1}$ and $\varepsilon > 0$, and let N be as in ineq 4.1. By Bernoulli's inequality,

$$\forall n \in \mathbb{N}_{\geq 1}: \quad \left(1+\frac{x}{n}\right)^n \geq 1+x.$$

For every $n \geq N$, therefore,

$$\exp(x) > \left(1+\frac{x}{n}\right)^n - \varepsilon \geq 1+x-\varepsilon.$$

Since $\varepsilon > 0$ is arbitrary,

$$e^x \geq 1+x.$$

If $x < -1$, then $1+x < 0 < e^x$, so the inequality holds there too.

Hence

$$\forall x \in \mathbb{R}: \quad e^x \geq 1+x. \tag{4.2}$$

Step 3: supporting-line criterion.

Substituting $z = x - y$ in ineq 4.2 and multiplying by $e^y > 0$ gives the supporting-line inequality

$$\forall x, y \in \mathbb{R} : \quad e^x \geq e^y + e^y(x - y).$$

By the supporting-line criterion (a differentiable f is convex if and only if its graph lies above each tangent line; see the proof of Theorem 3.3), exp is convex on $\mathbb{R}$.

The approximation

$$(1 + x/n)^n \to e^x$$

is illustrated in Figure 3.

□

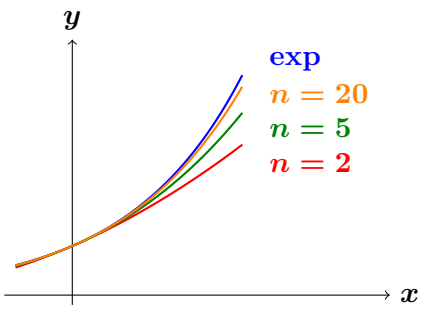


Figure 3: Bernoulli's approximation: for each $x \in \mathbb{R}$ and each $\varepsilon > 0$ there is N such that $\left|(1 + x/n)^n - \exp(x)\right| < \varepsilon$ for all $n \geq N$.

4.2. Inverse of the concave logarithm (Eighth proof)

Theorem 4.2.

exp is convex on $\mathbb{R}$ as the inverse of the concave logarithm.

Proof.

The logarithm

$$\log : \mathbb{R}_{>0} \to \mathbb{R},$$

is strictly concave, because

$$\log''(x) < 0.$$

The inverse of a strictly increasing concave function is convex. Since exp is the inverse of log, exp is convex.

□

4.3. Pointer to the Lean's proof (Ninth proof)

The Lean's library contains a proof of `strict_convex_on_exp` that is deliberately elementary and avoids the second-derivative test [2].

The file is on the path to Hölder and Minkowski, so it cannot invoke the second-derivative test.

The proof uses a slope argument instead.

We refer to Section 5 for the full mathematical adaptation.

5. The Lean's proof

5.1. Source code.

The Lean file `analysis/convex/specific_functions/basic.lean` (authors: Yury Kudryashov, Sébastien Gouëzel, Heather Macbeth. Copyright 2020, Apache 2.0) proves

$$\texttt{strict_convex_on_exp} : \texttt{strict_convex_on}\ \mathbb{R}\ \texttt{univ}\ \exp,$$

and derives

$$\texttt{convex_on_exp} : \texttt{convex_on}\ \mathbb{R}\ \texttt{univ}\ \exp,$$

by `strict_convex_on_exp.convex_on`.

The file header states that "the proofs in this file are deliberately elementary, not by appealing to the sign of the second derivative," comma because the file sits early in the import hierarchy.

The Lean's proof uses two auxiliary facts, which we now state and prove in ordinary mathematical language. (See [2])

5.2. Auxiliary lemma (strict supporting-line inequality)

Lemma 5.1 (Strict supporting-line inequality).
For every $t \in \mathbb{R}$ with $t \neq 0$,

$$e^t > 1 + t. \tag{5.1}$$

Proof 1: direct, via the power series.

If $t > 0$, then

$e^t > 1+t$ because for every $N \geq 2$, $\displaystyle\sum_{n=0}^{N} t^n/n! > 1+t$, and the inequality passes to $\exp(t)$,

and every term of the sum is strictly positive, so

$$\forall t \in \mathbb{R}_{>0} : \ e^t > 1 + t.$$

- If $-1 < t < 0$, write $t = -s$ with $0 < s < 1$.
  For each $k \geq 0$, the pair of consecutive terms

$$\frac{s^{2k}}{(2k)!} - \frac{s^{2k+1}}{(2k+1)!} = \frac{s^{2k}}{(2k)!}\left(1 - \frac{s}{2k+1}\right)$$

  is strictly positive because $s < 1 \leq 2k + 1$.
  For every $K \in \mathbb{N}$ the partial sum

$$\sum_{k=0}^{K} \left(s^{2k}/(2k)! - s^{2k+1}/(2k+1)!\right)$$

  exceeds $1 - s$ (the $k = 0$ term already contributes $1 - s$, every other term is positive), and the inequality passes to $\exp(-s)$.

- If $t \le -1$, then
$$1 + t \le 0 < e^t,$$
so the inequality holds trivially.

□

Proof 2: via the logarithm, as in Lean.

The power-series (or integral) definition of log gives
$$\forall x \in \mathbb{R}_{>0} : \ x \neq 1 \Rightarrow \log x < x - 1.$$
Apply this with $x = e^t$ for $t \neq 0$; then $x > 0$ and $x \neq 1$, so
$$\log(e^t) < e^t - 1,$$
i.e.
$$t < e^t - 1,$$
i.e.
$$1 + t < e^t.$$

□

Both proofs are elementary.

- Proof 2 mirrors the Lean derivation (where the strict inequality is obtained from the log inequality).
- Proof 1 stays within the power-series framework used throughout this note.

5.3. Auxiliary lemma (slope criterion)

Lemma 5.2 (Slope criterion for strict convexity).
Let $I \subseteq \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$.
The following statements are equivalent:
(1) f is strictly convex on I.
(2) for all $x < y < z$ in I,
$$\frac{f(y) - f(x)}{y - x} < \frac{f(z) - f(y)}{z - y}. \tag{5.2}$$

Proof.

For $x < y < z$, write
$$y = tx + (1 - t)z$$
with
$$t = (z - y)/(z - x) \in (0, 1).$$
Then
$$y - x = (1 - t)(z - x),$$
and
$$z - y = t(z - x).$$

Multiplying ineq 5.2 by $(z-x)>0$ and then by $t(1-t)>0$, it becomes

$$t\big(f(y)-f(x)\big)<(1-t)\big(f(z)-f(y)\big),$$

which rearranges to

$$f(y)<tf(x)+(1-t)f(z),$$

the strict convexity inequality.

□

5.4. Proof of the exp convexity (Lean slope argument)

Theorem 5.1 (Strict convexity of exp, Lean form).
exp is strictly convex on $\mathbb{R}$ by the slope criterion. Consequently, exp is convex on $\mathbb{R}$.

Proof.

By Lemma 5.2, it suffices to show that for all $x<y<z$ in $\mathbb{R}$,

$$\frac{e^y-e^x}{y-x}<\frac{e^z-e^y}{z-y}.$$

We prove the stronger chain

$$\forall x,y,z\in\mathbb{R}:\quad x<y<z\implies\frac{e^y-e^x}{y-x}<e^y<\frac{e^z-e^y}{z-y}. \tag{5.3}$$

Left inequality: Put

$$h:=y-x>0.$$

By multiplicativity,

$$e^y-e^x=e^y\big(1-e^{x-y}\big)=e^y\big(1-e^{-h}\big).$$

By Lemma 5.1 with $t=-h\neq 0$, we have

$$e^{-h}>1-h,$$

so

$$1-e^{-h}<h.$$

Hence

$$e^y-e^x<e^y\cdot h=e^y(y-x).$$

Dividing by $y-x>0$ gives

$$(e^y-e^x)/(y-x)<e^y.$$

Right inequality: Put

$$k:=z-y>0.$$

By multiplicativity,

$$e^z-e^y=e^y\big(e^{z-y}-1\big)=e^y\big(e^k-1\big).$$

By Lemma 5.1 with $t=k\neq 0$, we have

$$e^k>1+k,$$

so

$$e^k-1>k.$$

Hence

$$e^z - e^y > e^y \cdot k = e^y(z - y).$$

Dividing by $z - y > 0$ gives

$$e^y < (e^z - e^y)/(z - y).$$

Combining, ineq 5.3 holds, so the slopes of exp are strictly increasing.

By Lemma 5.2, exp is strictly convex on $\mathbb{R}$, and strict convexity implies convexity.
The chain of slopes in ineq 5.3 is drawn in Figure 4.

□

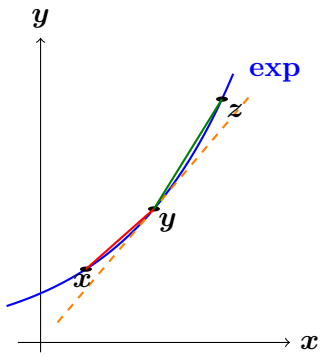


Figure 4: Slope criterion for the Lean proof: for $x < y < z$, the chord from x to y (red) has slope strictly less than the tangent at y (orange, slope e^y), which in turn is strictly less than the slope of the chord from y to z (green).

Remark 5.1 (Relation to Theorem 3.3).
Theorems 3.3 and 5.1 both rest on the supporting-line inequality.

The structural difference is that Theorem 3.3 uses the supporting-line (tangent) criterion (a differentiable f is convex iff its graph lies above each tangent line), which requires differentiability, whereas Theorem 5.1 uses the slope criterion (a function is strictly convex iff adjacent slopes are strictly increasing), which requires no differentiability and directly yields strict convexity.
The Lean's proof is in this sense more elementary than Theorem 3.3.
The price is the strict auxiliary inequality for $t \neq 0$

$$e^t > 1 + t$$

rather than the non-strict $e^t \geq 1 + t$.

Remark 5.2 (The remaining Lean content).
The same Lean file also proves, by the same elementary slope and algebraic methods, that:

- $x \mapsto x^n$ is convex on $\mathbb{R}_{>0}$ for every $n \in \mathbb{N}$, and on all of $\mathbb{R}$ when n is even.
- $x \mapsto x^m$ is convex on $\mathbb{R}_{>0}$ for every $m \in \mathbb{Z}$.
- log is strictly concave on $\mathbb{R}_{>0}$ and on $\mathbb{R}_{<0}$.
- $x \mapsto x^p$ is convex on $\mathbb{R}_{>0}$ for $p \geq 1$, and strictly convex for $p > 1$.

These are not needed for the convexity of exp, but they share the same design argument: "Elementary slope and algebraic arguments with no second-derivative test."

6. Reverse-mathematical classification proofs

Of the ten routes:

- Three are constructive in Bishop's sense (proofs $1, 2, 3$).
- Three are classical differential proofs (proofs $4, 5, 6$).
- Two are further classical or analytic proofs (proofs $7, 8$).
- Two concern the formal Lean's proof (proofs $9, 10$).

The constructive routes share the property that no appeal is made to the law of excluded middle, the mean value theorem, or the second-derivative test.
The classical routes are standard textbook proofs.
The formal routes exhibit the same underlying arguments in the language of the Lean theorem prover, with the slope criterion replacing the supporting-line criterion of the classical proof.

#	Mechanism	Reference	Constr.
1	Weighted AM–GM inequality	Theorem 2.1	Yes
2	Direct series on the non-negative reals, then translation	Theorems 2.2 and 2.3	Yes
3	Midpoint convexity via an algebraic square	Theorem 2.4	Yes
4	Second-derivative test	Theorem 3.1	No
5	Derivative strictly increasing	Theorem 3.2	No
6	Supporting line via the series	Theorem 3.3	No
7	Limit definition and Bernoulli	Theorem 4.1	No
8	Inverse of the concave logarithm	Theorem 4.2	No
9	Lean's formal proof (pointer)	§4	Formal
10	Lean's slope proof (detailed)	Theorem 5.1	Formal

The three constructive proofs are independent in their analytic inputs:

- Proof 1 uses the weighted AM–GM inequality.
- Proof 2 uses monomial convexity on $\mathbb{R}_+$.
- Proof 3 uses the algebraic identity $(\texttt{Real.exp}\,(x/2) - \texttt{Real.exp}\,(y/2))^2 \geq 0$.

They share only the power-series definition and its standard consequences. The two Lean's proofs share the slope criterion and the strict supporting-line inequality. Proof 10 in addition uses the multiplicativity of the exponential. Sections 2–5 prove the convexity of `Real.exp` via ten independent routes.
This section re-classifies each route in second-order arithmetic.
Entries carry a strength in $\{\mathsf{FOL}, \mathsf{RCA}_0, \mathsf{WKL}_0, \mathsf{ACA}_0, \mathsf{ZF}, \mathsf{ZFC}\}$.
No route uses $\mathsf{DC}\,(\mathbb{R})$ or any choice principle.

Ax 1	Recursive Comprehension Axiom	$\mathsf{RCA}_0 \;=\; \mathsf{FOL} + \Delta^0_1\text{-CA} + \Sigma^0_1\text{-Ind}$ $\Delta^0_1\text{-CA}\colon\ \exists X\,\forall n\,(n \in X \iff \varphi(n)),\quad \varphi \in \Delta^0_1$ $\Sigma^0_1\text{-Ind}\colon\ (\varphi(0) \wedge \forall n\,(\varphi(n) \to \varphi(n+1))) \to \forall n\,\varphi(n),\quad \varphi \in \Sigma^0_1$
Ax 2	Weak König's Lemma	$\mathsf{WKL}_0 \;=\; \mathsf{RCA}_0 + \mathsf{WKL}$ $\mathsf{WKL}:\ \forall T \subseteq \{0,1\}^{<\mathbb{N}},$ $\big(\forall n\,\exists \sigma \in T\,(\lvert\sigma\rvert \geq n) \to \exists f : \mathbb{N} \to \{0,1\}\,\forall n\,(f \restriction n \in T)\big)$
Ax 3	Arithmetical Comprehension Axiom	$\mathsf{ACA}_0 \;=\; \mathsf{RCA}_0 + \text{Arithmetical CA}$ Arithmetical CA: $\exists X\,\forall n\,(n \in X \iff \varphi(n))$, φ Arithmetical
Ax 4	Dependent Choice on the reals	$\mathsf{ZF} + \mathsf{DC}(\mathbb{R})\colon\ \forall A \subseteq \mathbb{R},\ A \neq \emptyset,\ \forall R \subseteq A \times A$ $\big(\forall x \in A\,\exists y \in A\,(x\,R\,y) \ \to$ $\forall a \in A\,\exists (a_n)\,(a_0 = a \wedge \forall n\,(a_n\,R\,a_{n+1}))\big)$
Ax 5	Axiom of Choice	$\mathsf{ZF} + \mathsf{AC}:\ \forall I \neq \emptyset\,\forall (A_i)_{i\in I}$ $\big(\forall i \in I\,(A_i \neq \emptyset) \ \to \exists f : I \to \bigcup_{i\in I} A_i\,\forall i \in I\,(f(i) \in A_i)\big)$ i.e. $\mathsf{ZFC} = \mathsf{ZF} + \mathsf{AC}$

The table below records the abbreviations used in the axiom entries above and throughout §6.

$\mathsf{FOL} :=$ first-order logic	quantifiers over individuals
$\mathrm{CA} := \exists X\,\forall n\,(n \in X \iff \varphi(n))$	comprehension scheme
$\mathrm{Ind} := \varphi(0) \wedge \forall n\,(\varphi(n) \to \varphi(n+1)) \to \forall n\,\varphi(n)$	induction scheme
Δ^0_1-CA	with $\varphi \in \Delta^0_1$
Σ^0_1-Ind	with $\varphi \in \Sigma^0_1$
arithmetical CA	with φ arithmetical
$\mathsf{RCA}_0 := \mathsf{FOL} + \Delta^0_1\text{-CA} + \Sigma^0_1\text{-Ind}$	system
$\mathsf{WKL} := \forall n\,\exists \sigma \in T\,(\lvert\sigma\rvert \geq n) \to \exists f\,\forall n\,(f \restriction n \in T)$	weak König's lemma
$\mathsf{WKL}_0 := \mathsf{RCA}_0 + \mathsf{WKL}$	system
$\mathsf{ACA}_0 := \mathsf{RCA}_0 +$ arithmetical CA	system
$\mathsf{DC}\,(\mathbb{R})$: ax:dc	dependent choice on $\mathbb{R}$
AC: axi:ac	axiom of choice
ZF: Zermelo–Fraenkel	set theory with $\in$
$\mathsf{ZFC} := \mathsf{ZF} + \mathsf{AC}$	with choice
$\mathbb{N} := \{0, 1, 2, \dots\}$	natural numbers
$f \restriction n := (f(0), \dots, f(n-1))$	restriction of f to n
$\{0,1\}^{<\mathbb{N}} := \bigcup_{n\in\mathbb{N}} \{0,1\}^n$	finite binary strings

6.1. Coding conventions

We adopt Bishop's Cauchy real with modulus coding: a real is a Cauchy sequence of rationals together with an explicit modulus of convergence.

Algebraic operations are primitive recursive in the codes.

A function

$$f : I \to \mathbb{R}$$

is coded by its graph plus a modulus of uniform continuity on each compact subinterval.

We write $\mathcal{N}(x)$ for the Lean neighbourhood filter, i.e. $\mathcal{N}(x)$ abbreviates `nhds x`. In Lean's notation the power series is

$$\forall x : \mathbb{R},\ \texttt{Real.exp}\, x = L \iff \forall \varepsilon > 0,\ \exists N,\ \forall m \geq N : \left| L - \sum_{n=0}^{m} x^n / (\mathsf{Nat.factorial}\, n : \mathbb{R}) \right| < \varepsilon.$$

This is available in RCA_0 because the partial sums form a Cauchy sequence with modulus dominated by $2^n/n!$ on $|x| \leq n$.
Multiplicativity:

$$\forall x\, y : \mathbb{R},\ \texttt{Real.exp}\ (x + y) = \texttt{Real.exp}\ x \cdot \texttt{Real.exp}\ y$$

follows from the binomial theorem plus Σ^0_1-comprehension. Convexity on I is the second-order formula

$$\begin{aligned} \mathsf{Conv}(f, I) : \iff\ & \forall x \in I,\ \forall y \in I,\ \forall t : \mathbb{R},\ 0 \leq t \to t \leq 1 \to \\ & f\big(t \cdot x + (1 - t) \cdot y\big) \leq t \cdot f(x) + (1 - t) \cdot f(y). \end{aligned}$$

6.2. Field and order axioms

The ordered-field and completeness axioms of $\mathbb{R}$, are in Lean's notation. All are ZF-realizable. None uses choice. Under the Bishop–Cauchy coding they are theorems of RCA_0.

Ax 6	$\forall x\, y\, z : \mathbb{R},\ (x + y) + z = x + (y + z)$	Add.Assoc.	ZF
Ax 7	$\forall x\, y : \mathbb{R},\ x + y = y + x$	Add.Comm.	ZF
Ax 8	$\forall x : \mathbb{R},\ 0 + x = x$	Add.Id.	ZF
Ax 9	$\forall x : \mathbb{R},\ x + (-x) = 0$	Add.Inv.	ZF
Ax 10	$\forall x\, y\, z : \mathbb{R},\ (x \cdot y) \cdot z = x \cdot (y \cdot z)$	Mul.Assoc.	ZF
Ax 11	$\forall x\, y : \mathbb{R},\ x \cdot y = y \cdot x$	Mul.Comm.	ZF
Ax 12	$\forall x : \mathbb{R},\ 1 \cdot x = x$	Mul.Id.	ZF
Ax 13	$\forall x : \mathbb{R},\ x \neq 0 \to x \cdot x^{-1} = 1$	Mul.Inv.	ZF
Ax 14	$\forall x\, y\, z : \mathbb{R},\ x \cdot (y + z) = x \cdot y + x \cdot z$	Distrib.	ZF
Ax 15	$(0 : \mathbb{R}) \neq 1$	Nontriv.	ZF
Ax 16	$(\forall x : \mathbb{R},\ x \leq x)$ $\wedge (\forall x\, y : \mathbb{R},\ x \leq y \to y \leq x \to x = y)$ $\wedge (\forall x\, y\, z : \mathbb{R},\ x \leq y \to y \leq z \to x \leq z)$ $\wedge (\forall x\, y : \mathbb{R},\ x \leq y \vee y \leq x)$	LinOrder	ZF
Ax 17	$\forall x\, y\, z : \mathbb{R},\ x \leq y \to x + z \leq y + z$	AddMon	ZF
Ax 18	$\forall x\, y : \mathbb{R},\ 0 \leq x \to 0 \leq y \to 0 \leq x \cdot y$	MulMon	ZF
Ax 19	$\forall S : \mathsf{Set}\ \mathbb{R},\ (\exists x,\ x \in S) \to$ $(\exists M,\ \forall x \in S,\ x \leq M) \to$ $\exists s,\ \mathsf{IsLUB}\ S\ s$	Cut	ZF

The Cut axiom enters no convexity proof in this section: The series for `Real.exp` converges with an explicit modulus, so no completeness is needed. All ten routes are therefore Cut-free.

6.3. Postulates

Internal statements, each tagged with its strength over RCA_0, in Lean's notation.

- $(\cdot : \mathbb{R})$ denotes the coercion $\mathbb{Q} \to \mathbb{R}$.
- Routes 1, 2, 3, 6, 7, 8, 9, and 10 use only RCA_0 entries.
- Routes 4 and 5 invoke the general second-derivative test and the monotone-derivative criterion.

Those, together with the two FTC tools they depend on (**conv-FTC** and **conv-int-strict**), are WKL_0 over RCA_0 (see [5, II.7, IV.2.3]), so Routes 4 and 5 are classified as WKL_0. The classical mean value theorem (**MVT**) is recorded as a classical reference only and is not used directly by any route.

The postulate **ExpPow** is stated for real t. Its RCA_0-derivation parallels the rational-to-real passage used in Lemma 2.1: the rational case

$$\exp((m/n)x) = (\exp x)^{m/n}$$

follows from multiplicativity (**ExpMult**) and the existence of n-th roots, and the passage to real t is by continuity of

$$t \mapsto \exp(tx)$$
$$t \mapsto (\exp x)^t,$$

both available in RCA_0 from the power-series modulus.

It is recorded as a single postulate only for brevity; its status is that of a derived RCA_0-postulate, exactly as

$$\textbf{AMGM}\,(\mathbb{R}) \longleftarrow \textbf{AMGM}\,(\mathbb{Q})\,.$$

Pos 1	$\forall x : \mathbb{R},\ \exists M : \mathbb{N},\ \lvert x\rvert \le (M : \mathbb{R})$	**Arch**	RCA_0
Pos 2	$\forall a\, b : \mathbb{R},\ 0 < a \to 0 < b \to$ $\forall t : \mathbb{Q},\ 0 \le t \to t \le 1 \to$ $\texttt{Real.rpow}\ a\ (t : \mathbb{R}) \cdot \texttt{Real.rpow}\ b\ (1 - (t : \mathbb{R}))$ $\le (t : \mathbb{R}) \cdot a + (1 - (t : \mathbb{R})) \cdot b$	**AMGM** $(\mathbb{Q})$	RCA_0
Pos 3	$\forall a\, b : \mathbb{R},\ 0 < a \to 0 < b \to$ $\forall t : \mathbb{R},\ 0 \le t \to t \le 1 \to$ $\texttt{Real.rpow}\ a\ t \cdot \texttt{Real.rpow}\ b\ (1 - t) \le t \cdot a + (1 - t) \cdot b$	**AMGM** $(\mathbb{R})$	RCA_0
Pos 4	$\forall x : \mathbb{R},\ \forall \varepsilon : \mathbb{R},\ 0 < \varepsilon \to$ $\exists N : \mathbb{N},\ \forall m\, n : \mathbb{N},\ N \le n \to n \le m \to$ $\left\lvert \sum_{k \in \mathsf{Finset.Icc}\ n\ m} x^k / (\mathsf{Nat.factorial}\ k : \mathbb{R}) \right\rvert < \varepsilon$	**SerCauchy**	RCA_0
Pos 5	$\forall x\, y : \mathbb{R},\ \texttt{Real.exp}\ (x + y) = \texttt{Real.exp}\ x \cdot \texttt{Real.exp}\ y$	**ExpMult**	RCA_0
Pos 6	$\forall x : \mathbb{R},\ \forall t : \mathbb{R},\ 0 \le t \to t \le 1 \to$ $\texttt{Real.exp}\ (t \cdot x) = \texttt{Real.rpow}\ (\texttt{Real.exp}\ x)\ t$	**ExpPow**	RCA_0
Pos 7	$\forall n : \mathbb{N},\ 1 \le n \to$ $\forall x\, y : \mathbb{R},\ 0 \le x \to 0 \le y \to$ $\forall t : \mathbb{R},\ 0 \le t \to t \le 1 \to$ $(t \cdot x + (1 - t) \cdot y)^n \le t \cdot x^n + (1 - t) \cdot y^n$	**Monom.**	RCA_0
Pos 8	$\forall x\, y : \mathbb{R},\ \texttt{Real.exp}\ ((x + y)/2) \le (\texttt{Real.exp}\ x + \texttt{Real.exp}\ y)/2$	**Midpt**	RCA_0

Pos 9	$\forall f : \mathbb{R} \to \mathbb{R},$ MidpointConvexOn $\mathbb{R}$ Set.univ $f \to$ Continuous $f \to$ ConvexOn $\mathbb{R}$ Set.univ f	**Mid2Conv**	**RCA$_0$**
Pos 10	$\forall f : \mathbb{R} \to \mathbb{R},\ \forall a\ b : \mathbb{R},\ a < b \to$ ContinuousOn f (Set.Icc $a\ b$) $\to$ DifferentiableOn $\mathbb{R}\ f$ (Set.Ioo $a\ b$) $\to$ ContinuousOn (deriv f) (Set.Icc $a\ b$) $\to$ $\int_a^b$ deriv $f\ t\,dt = f\ b - f\ a$	**FTC**	**WKL$_0$**
Pos 11	$\forall g\ h : \mathbb{R} \to \mathbb{R},\ \forall a\ b : \mathbb{R},\ a < b \to$ ContinuousOn g (Set.Icc $a\ b$) $\to$ ContinuousOn h (Set.Icc $a\ b$) $\to$ $(\forall t \in$ Set.Ioo $a\ b,\ g\ t < h\ t) \to$ $\int_a^b g\ t\,dt < \int_a^b h\ t\,dt$	**IntStr**	**WKL$_0$**
Pos 12	$\forall f : \mathbb{R} \to \mathbb{R},$ ContDiff $\mathbb{R}\ 2\ f \to$ $(\forall x : \mathbb{R},$ deriv (deriv f) $x > 0) \to$ StrictMono (deriv f)	**DerPosMono**	**WKL$_0$**
Pos 13	$\forall f : \mathbb{R} \to \mathbb{R},\ \forall a\ b : \mathbb{R},\ a < b \to$ ContinuousOn f (Set.Icc $a\ b$) $\to$ DifferentiableOn $\mathbb{R}\ f$ (Set.Ioo $a\ b$) $\to$ $\exists c \in$ Set.Ioo $a\ b,$ deriv $f\ c = (f\ b - f\ a)/(b - a)$	**MVT**	**WKL$_0$**
Pos 14	$\forall f : \mathbb{R} \to \mathbb{R},$ Continuous $f \to$ Differentiable $\mathbb{R}\ f \to$ Monotone (deriv f) $\to$ ConvexOn $\mathbb{R}$ Set.univ f	**DerMon**	**WKL$_0$**
Pos 15	$\forall f : \mathbb{R} \to \mathbb{R},\ (\forall x\ y : \mathbb{R},\ f\ x \geq f\ y +$ deriv $f\ y \cdot (x - y)) \to$ ConvexOn $\mathbb{R}$ Set.univ f	**Support**	**RCA$_0$**
Pos 16	$\forall z : \mathbb{R},$ Real.exp $z \geq 1 + z$	**EZ**	**RCA$_0$**
Pos 17	$\forall x : \mathbb{R},\ -1 \leq x \to$ $\forall n : \mathbb{N},\ 1 \leq n \to (1 + x/(n : \mathbb{R}))^n \geq 1 + x$	**Bern**	**RCA$_0$**
Pos 18	StrictConcaveOn $\mathbb{R}$ (Set.Ioi 0) Real.log	**LogConc**	**RCA$_0$**
Pos 19	$\forall f\ g : \mathbb{R} \to \mathbb{R},$ Set.InvOn $g\ f$ (Set.Ioi 0) $(f$ '' Set.Ioi $0) \to$ StrictMonoOn f (Set.Ioi 0) $\to$ StrictConcaveOn $\mathbb{R}$ (Set.Ioi 0) $f \to$ StrictConvexOn $\mathbb{R}$ $(f$ '' Set.Ioi $0)\ g$	**LogInv**	**RCA$_0$**
Pos 20	$\forall t : \mathbb{R},\ t \neq 0 \to$ Real.exp $t > 1 + t$	**StrSupp**	**RCA$_0$**
Pos 21	$\forall f : \mathbb{R} \to \mathbb{R},$ StrictConvexOn $\mathbb{R}$ Set.univ $f \leftrightarrow$ $\forall x\ y\ z : \mathbb{R},\ x < y \to y < z \to$ $(f\ y - f\ x)/(y - x) < (f\ z - f\ y)/(z - y)$	**SlopeCrit**	**RCA$_0$**
Pos 22	$\forall f : \mathbb{R} \to \mathbb{R},\ \forall x : \mathbb{R},\ \forall u : \mathbb{N} \to \mathbb{R},$ ContinuousAt $f\ x \to$ Filter.Tendsto u Filter.atTop $(\mathcal{N}\ x) \to$ Filter.Tendsto $(\lambda n,\ f\ (u\ n))$ Filter.atTop $(\mathcal{N}\ (f\ x))$	**SeqCont**	**RCA$_0$**
Pos 23	$\forall x : \mathbb{R},$ Real.exp $x > 0$	**ExpPos**	**RCA$_0$**
Pos 24	$\forall x : \mathbb{R},$ deriv Real.exp $x =$ Real.exp x	**ExpDeriv**	**RCA$_0$**
Pos 25	Continuous Real.exp	**ExpCont**	**RCA$_0$**
Pos 26	ContDiff $\mathbb{R}\ 2$ Real.exp	**ExpC2**	**RCA$_0$**
Pos 27	$\forall f : \mathbb{R} \to \mathbb{R},$ ContDiff $\mathbb{R}\ 2\ f \to$ $(\forall x : \mathbb{R},$ deriv (deriv f) $x > 0) \to$ StrictConvexOn $\mathbb{R}$ Set.univ f	**Conv2nd**	**WKL$_0$**
Pos 28	$\forall f : \mathbb{R} \to \mathbb{R},$ ContDiff $\mathbb{R}\ 1\ f \to$ StrictMono (deriv f) $\to$ StrictConvexOn $\mathbb{R}$ Set.univ f	**ConvDerStr**	**WKL$_0$**

Pos 29	$\forall x \in \mathbb{R},\ \forall \varepsilon \in \mathbb{R}_{>0},\ \exists N \in \mathbb{N},\ \forall n \in \mathbb{N}:\quad n \geq N \Rightarrow \left\lvert (1 + x/(n:\mathbb{R}))^n - \texttt{Real.exp}\ x \right\rvert < \varepsilon$	**LimitExp**	**RCA$_0$**
Pos 30	$\forall x : \mathbb{R},\ \texttt{Real.log}\ (\texttt{Real.exp}\ x) = x \wedge (\forall x : \mathbb{R},\ 0 < x \to \texttt{Real.exp}\ (\texttt{Real.log}\ x) = x)$	**ExpLogEq**	**RCA$_0$**
Pos 31	**StrictMonoOn** `Real.log` (**Set.Ioi** 0)	**LogMono**	**RCA$_0$**

6.4. Definitions

Definitional abbreviations available in RCA$_0$:
Set notation by Δ_1^0-comprehension: Conv, StrictConv, Cont by Σ_1^0 predicates. `Real.exp`, `Real.log` **by unique existence + graphs.**

Def 1	`Real.exp` $x :=$ the unique L such that $\forall \varepsilon \in \mathbb{R}_{>0}\ \exists N \in \mathbb{N}\ \forall m \geq N:\ \left\lvert L - \sum_{k=0}^{m} x^k/k! \right\rvert < \varepsilon$	**exp**	**RCA$_0$**
Def 2	**Conv**$(f, I) :\iff \forall x \in I,\ \forall y \in I,\ \forall t : \mathbb{R},\ 0 \leq t \to t \leq 1 \to f(t \cdot x + (1-t) \cdot y) \leq t \cdot f(x) + (1-t) \cdot f(y)$	**Conv**	**RCA$_0$**
Def 3	**StrictConv**$(f, I) :\iff$ **Conv**$(f, I) \wedge (\forall x \in I,\ \forall y \in I,\ x \neq y \to \forall t : \mathbb{R},\ 0 < t \to t < 1 \to f(t \cdot x + (1-t) \cdot y) < t \cdot f(x) + (1-t) \cdot f(y))$	**StrConv**	**RCA$_0$**
Def 4	**Cont**$(f, I) :\iff \forall a\ b : \mathbb{R},\ a < b \to \forall \varepsilon : \mathbb{R},\ 0 < \varepsilon \to \exists \delta : \mathbb{R},\ 0 < \delta \wedge \forall x \in I \cap$ **Set.Icc** $a\ b,\ \forall y \in I \cap$ **Set.Icc** $a\ b,\ |x - y| < \delta \to |f\ x - f\ y| < \varepsilon$	**Cont**	**RCA$_0$**
Def 5	**Midpoint**$(f, I) :\iff \forall x \in I,\ \forall y \in I,\ f((x+y)/2) \leq (f\ x + f\ y)/2$	**Midpt**	**RCA$_0$**
Def 6	**Slope**$(f, x, y) := (f\ y - f\ x)/(y - x)$ for $x \neq y$	**Slope**	**RCA$_0$**
Def 7	**Diff**$(f, x) :\iff \exists L,$ **Filter.Tendsto** $(\lambda h,\ (f\ (x+h) - f\ x)/h)\ (\mathcal{N}\ 0)\ (\mathcal{N}\ L)$ and then **deriv** $f\ x :=$ the unique such L	**Diff**	**RCA$_0$**
Def 8	$C^k(I)(f) :\iff \exists g_0, \ldots, g_k,\ g_0 = f \wedge \forall j < k,\ g_{j+1} =$ **deriv** g_j	**C^k**	**RCA$_0$**
Def 9	`Real.log` $x :=$ the unique y such that `Real.exp` $y = x \quad (x > 0)$; the identity `Real.log` $x = \int_1^x dt/t$ is a consequence of the FTC, hence **WKL$_0$**-provable, and is not used as a definitional basis. Existence of y from Pos:1 + monotone IVT; uniqueness from Pos:20.	**log**	**RCA$_0$**
Def 10	if `Real.log` is constructed first, `Real.exp` is characterised by `Real.log` $\circ$ `Real.exp` $=$ **id** and $\forall x,\ 0 < x \to$ `Real.exp` (`Real.log` $x) = x$	**lninv**	**RCA$_0$**

6.5. Statements (the ten routes re-classified)

Every route appears below as a tagged entry. Routes 9 and 10 carry separate statement entries: Route 9 is the one-line pointer to the Lean's source, Route 10 is the detailed slope-criterion derivation. Both resolve to the same underlying theorem.

Thm 1	$\forall x\, y : \mathbb{R},\ \forall t : \mathbb{R},\ 0 \le t \to t \le 1 \to$ Real.exp $(t \cdot x + (1-t) \cdot y) \le t \cdot$ Real.exp $x + (1-t) \cdot$ Real.exp y (via weighted AM–GM)	AMGM	RCA_0
Thm 2	$\forall x\, y : \mathbb{R},\ 0 \le x \to 0 \le y \to \forall t : \mathbb{R},\ 0 \le t \to t \le 1 \to$ Real.exp $(t \cdot x + (1-t) \cdot y) \le t \cdot$ Real.exp $x + (1-t) \cdot$ Real.exp y (direct series on $\mathbb{R}_{\ge 0}$)	Ser	RCA_0
Thm 3	series convexity on $\mathbb{R}_{\ge 0} \to$ convexity on $\mathbb{R}$ (translation)	Trans	RCA_0
Thm 4	MidpointConvexOn $\mathbb{R}$ Set.univ Real.exp $\to$ Continuous Real.exp $\to$ ConvexOn $\mathbb{R}$ Set.univ Real.exp	Mid	RCA_0
Thm 5	$(\forall x,$ deriv (deriv Real.exp) $x > 0) \to$ StrictConvexOn $\mathbb{R}$ Set.univ Real.exp	2nd	WKL_0
Thm 6	StrictMono (deriv Real.exp) $\to$ StrictConvexOn $\mathbb{R}$ Set.univ Real.exp	Deriv	WKL_0
Thm 7	$(\forall x\, y : \mathbb{R},$ Real.exp $x \ge$ Real.exp $y +$ Real.exp $y \cdot (x - y)) \to$ ConvexOn $\mathbb{R}$ Set.univ Real.exp	Supp	RCA_0
Thm 8	$(\forall x : \mathbb{R},$ Filter.Tendsto $(\lambda n : \mathbb{N},\ (1 + x/(n : \mathbb{R}))^n)$ Filter.atTop $(\mathcal{N}$ (Real.exp $x))) \to (\forall x : \mathbb{R},\ -1 \le x \to \forall n : \mathbb{N},\ 1 \le n \to (1 + x/(n : \mathbb{R}))^n \ge 1 + x) \to$ ConvexOn $\mathbb{R}$ Set.univ Real.exp	Bern	RCA_0
Thm 9	StrictConcaveOn $\mathbb{R}$ (Set.Ioi 0) Real.log $\to$ ConvexOn $\mathbb{R}$ Set.univ Real.exp	Log	RCA_0
Thm 10	Real.exp strictly convex via slope criterion (Lean pointer)	Lean	RCA_0
Thm 11	Real.exp strictly convex via slope criterion (detailed Lean form)	LeanDet	RCA_0

Routes 1, 2, 3, 6, 7, 8, 9, 10 are RCA_0-provable under Bishop–Cauchy coding. Routes 4 and 5 are WKL_0-provable because they invoke the FTC/2nd-derivative criterion. The theorem itself (convexity of Real.exp) is therefore RCA_0-provable, by any of Routes 1, 2, 3, 6, 7, 8, 9, or 10. None of the routes uses ACA_0, $\mathsf{DC}\,(\mathbb{R})$, or any choice principle.

6.6. Proofs (dependency tables)

One starred subsection per route, in route order. Each proof is a four-column dependency table: first row **Hyp**, last row $\to$ **I**$(1, \ldots, N-1)$. Justifications use only **Hyp**, **def**, **Ax**, **Pos**, **Def**, **Lem**, **Thm**, **Eq**$(\ldots)$, **Midpoint**, **SlopeCrit**, **ExpMult**, **FTC**, **IntStr**, **Support**, **Series**, **Bernoulli**, **LogConc**, **LogInv**, **Slope**, **Ref**, or a logical inference rule

$$(\to \mathsf{I}, \wedge\mathsf{E}, \vee\mathsf{E}, \forall\mathsf{I}, \forall\mathsf{E}, \exists\mathsf{I}, \exists\mathsf{E}, \mathsf{RAA}).$$

No external citation appears inside a proof table.

Proof of Theorem 1 (Route 1: weighted AM–GM)

1	$x\, y : \mathbb{R},\ t : \mathbb{R},\ 0 \le t \to t \le 1$	hypothesis	Hyp
2	Real.exp $(t \cdot x + (1-t) \cdot y) =$ Real.rpow (Real.exp x) $t \cdot$ Real.rpow (Real.exp y) $(1-t)$	mult	Pos:5(1) $\wedge$ Pos:6(1)

3	$\texttt{Real.rpow}\ (\texttt{Real.exp}\ x)\ t \cdot \texttt{Real.rpow}\ (\texttt{Real.exp}\ y)\ (1-t) \le t \cdot \texttt{Real.exp}\ x + (1-t) \cdot \texttt{Real.exp}\ y$	AMGM	Pos:3(1)
4	$\texttt{Real.exp}\ (t \cdot x + (1-t) \cdot y) \le t \cdot \texttt{Real.exp}\ x + (1-t) \cdot \texttt{Real.exp}\ y$	combine	Eq(2, 3)
5	$\top$	close	$\to$ I(1, . . . , 4)

Proof of Theorem 2 (Route 2a: direct series on $\mathbb{R}_{\ge 0}$)

1	$x\ y : \mathbb{R},\ 0 \le x \to 0 \le y \to t : \mathbb{R},\ 0 \le t \to t \le 1$	hypothesis	Hyp
2	$\forall n : \mathbb{N},\ 1 \le n \to (t \cdot x + (1-t) \cdot y)^n \le t \cdot x^n + (1-t) \cdot y^n$	Monom	Pos:7(1)
3	$(t \cdot x + (1-t) \cdot y)^n / (\text{Nat.factorial}\ n : \mathbb{R}) \le t \cdot (x^n / (\text{Nat.factorial}\ n : \mathbb{R})) + (1-t) \cdot (y^n / (\text{Nat.factorial}\ n : \mathbb{R}))$	div	Eq(2)
4	$\sum_{n:\mathbb{N}} (t \cdot x + (1-t) \cdot y)^n / (\text{Nat.factorial}\ n : \mathbb{R}) \le t \cdot \sum_{n:\mathbb{N}} x^n / (\text{Nat.factorial}\ n : \mathbb{R}) + (1-t) \cdot \sum_{n:\mathbb{N}} y^n / (\text{Nat.factorial}\ n : \mathbb{R})$	sum	Eq(3) $\wedge$ Pos:4
5	$\texttt{Real.exp}\ (t \cdot x + (1-t) \cdot y) \le t \cdot \texttt{Real.exp}\ x + (1-t) \cdot \texttt{Real.exp}\ y$	def	Eq(4) $\wedge$ Def:1
6	$\top$	close	$\to$ I(1, . . . , 5)

Proof of Theorem 3 (Route 2b: translation to all of $\mathbb{R}$)

1	$x\ y : \mathbb{R},\ t : \mathbb{R},\ 0 \le t \to t \le 1$	hypothesis	Hyp
2	$\exists M_x : \mathbb{N},\ \lvert x\rvert \le (M_x : \mathbb{R}),\quad \exists M_y : \mathbb{N},\ \lvert y\rvert \le (M_y : \mathbb{R})$	Arch	Pos:1(1)
3	$\text{let}\ M := \max\{M_x, M_y\};\quad \exists M : \mathbb{N},\ \lvert x\rvert \le (M : \mathbb{R}) \wedge \lvert y\rvert \le (M : \mathbb{R})$	max	Eq(2)
4	$x' := x + (M : \mathbb{R}),\ y' := y + (M : \mathbb{R}),\ 0 \le x' \wedge 0 \le y'$	shift	Eq(1, 3)
5	$t \cdot x + (1-t) \cdot y = t \cdot x' + (1-t) \cdot y' - (M : \mathbb{R})$	algebra	Eq(4)
6	$\texttt{Real.exp}\ (t \cdot x + (1-t) \cdot y) = \texttt{Real.exp}\ (t \cdot x' + (1-t) \cdot y') \cdot \texttt{Real.exp}\ (-(M : \mathbb{R}))$	mult	Eq(5) $\wedge$ Pos:5(5)
7	$\texttt{Real.exp}\ (t \cdot x' + (1-t) \cdot y') \le t \cdot \texttt{Real.exp}\ x' + (1-t) \cdot \texttt{Real.exp}\ y'$	Thm	Thm:2(4)
8	$\texttt{Real.exp}\ (-(M : \mathbb{R})) > 0$	positivity	Pos:23(1)
9	$\texttt{Real.exp}\ (t \cdot x' + (1-t) \cdot y') \cdot \texttt{Real.exp}\ (-(M : \mathbb{R})) \le (t \cdot \texttt{Real.exp}\ x' + (1-t) \cdot \texttt{Real.exp}\ y') \cdot \texttt{Real.exp}\ (-(M : \mathbb{R}))$	mono	Eq(7, 8)
10	$t \cdot \texttt{Real.exp}\ x' \cdot \texttt{Real.exp}\ (-(M : \mathbb{R})) = t \cdot \texttt{Real.exp}\ x,\quad (1-t) \cdot \texttt{Real.exp}\ y' \cdot \texttt{Real.exp}\ (-(M : \mathbb{R})) = (1-t) \cdot \texttt{Real.exp}\ y$	mult	Eq(4, 6) $\wedge$ Pos:5
11	$(t \cdot \texttt{Real.exp}\ x' + (1-t) \cdot \texttt{Real.exp}\ y') \cdot \texttt{Real.exp}\ (-(M : \mathbb{R})) = t \cdot \texttt{Real.exp}\ x + (1-t) \cdot \texttt{Real.exp}\ y$	distribute	Eq(10)
12	$\texttt{Real.exp}\ (t \cdot x + (1-t) \cdot y) \le t \cdot \texttt{Real.exp}\ x + (1-t) \cdot \texttt{Real.exp}\ y$	chain	Eq(6, 9, 11)
13	$\top$	close	$\to$ I(1, . . . , 12)

Proof of Theorem 4 (Route 3: midpoint)

1	$x\ y : \mathbb{R}$	hypothesis	Hyp
2	$(\texttt{Real.exp}\ (x/2) - \texttt{Real.exp}\ (y/2))^2 \ge 0$	square	Ax:18 $\wedge$ Ax:16
3	$\texttt{Real.exp}\ x + \texttt{Real.exp}\ y \ge 2 \cdot \texttt{Real.exp}\ ((x+y)/2)$	expand	Eq(2) $\wedge$ Pos:5
4	$\texttt{Real.exp}\ ((x+y)/2) \le (\texttt{Real.exp}\ x + \texttt{Real.exp}\ y)/2$	divide	Eq(3)

5	**Midpoint** `Real.exp`	def	Def:5(4)
6	**Continuous** `Real.exp`	series	Pos:25(1)
7	**ConvexOn** $\mathbb{R}$ **Set.univ** `Real.exp`	cont	Pos:9(5, 6)
8	$\top$	close	$\to$ I(1, . . . , 7)

Proof of Theorem 5 (Route 4: second derivative)

1	**ContDiff** $\mathbb{R}$ 2 `Real.exp`	series	Pos:26
2	**deriv** `Real.exp` x = `Real.exp` x	deriv	Pos:24(1)
3	$\forall x : \mathbb{R}$, `Real.exp` $x > 0$	pos	Pos:23(1)
4	$\forall x : \mathbb{R}$, **deriv** (**deriv** `Real.exp`) x = `Real.exp` $x > 0$	combine	Eq(2, 3)
5	**StrictConvexOn** $\mathbb{R}$ **Set.univ** `Real.exp`	2nd	Pos:27(1, 4)
6	$\top$	close	$\to$ I(1, . . . , 5)

Proof of Theorem 6 (Route 5: derivative strictly increasing)

1	**ContDiff** $\mathbb{R}$ 2 `Real.exp`	series	Pos:26
2	$\forall x : \mathbb{R}$, **deriv** `Real.exp` x = `Real.exp` x	deriv	Pos:24
3	$\forall x : \mathbb{R}$, `Real.exp` $x > 0$	pos	Pos:23
4	$\forall x : \mathbb{R}$, **deriv** (**deriv** `Real.exp`) x = `Real.exp` $x > 0$	combine	Eq(2, 3)
5	**StrictMono** (**deriv** `Real.exp`)	FTC	Pos:12(1,4)
6	**ContDiff** $\mathbb{R}$ 1 `Real.exp`	def	Def:8(1)
7	**StrictConvexOn** $\mathbb{R}$ **Set.univ** `Real.exp`	deriv $-$ mon	Pos:28(6,5)
8	$\top$	close	$\to$ I(1, . . . , 7)

Proof of Theorem 7 (Route 6: supporting line)

1	$\forall z : \mathbb{R}$, `Real.exp` $z \geq 1 + z$	EZ	Pos:16
2	$\forall x\, y : \mathbb{R}$, `Real.exp` $(x - y) \geq 1 + (x - y)$	substitute	Eq(1)
3	$\forall x\, y : \mathbb{R}$, `Real.exp` $x \geq$ `Real.exp` $y +$ `Real.exp` $y \cdot (x - y)$	mult	Eq(2) $\wedge$ Pos:5 $\wedge$ Pos:23
4	**ConvexOn** $\mathbb{R}$ **Set.univ** `Real.exp`	support	Pos:15(3)
5	$\top$	close	$\to$ I(1, . . . , 4)

Proof of Theorem 8 (Route 7: Bernoulli)

1	$\forall x \in \mathbb{R}$, $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$, $\forall n \in \mathbb{N}$: $n \geq N \Rightarrow \lvert(1 + x/n)^n - \exp(x)\rvert < \varepsilon$	def $-$ $\lim_{\varepsilon\text{-}N}$	Pos:29
2	$\forall x : \mathbb{R}$, $-1 \leq x \to \forall n : \mathbb{N}$, $1 \leq n \to (1 + x/(n : \mathbb{R}))^n \geq 1 + x$	Bernoulli	Pos:17(1)
3	$\forall x : \mathbb{R}$, $-1 \leq x \to$ `Real.exp` $x \geq 1 + x$	$\lim_{\varepsilon\text{-}N}$	Eq(1, 2)
4	$\forall x : \mathbb{R}$, $x < -1 \to$ `Real.exp` $x \geq 1 + x$	trivial	Pos:23
5	$\forall x : \mathbb{R}$, `Real.exp` $x \geq 1 + x$	cases	Eq(3, 4)

6	$\forall x\ y : \mathbb{R},\ \texttt{Real.exp}\ x \geq \texttt{Real.exp}\ y + \texttt{Real.exp}\ y \cdot (x - y)$	sub + mult	Eq(5) ∧ Pos:5 ∧ Pos:23
7	ConvexOn $\mathbb{R}$ Set.univ Real.exp	support	Pos:15(6)
8	$\top$	close	→ I(1, . . . , 7)

Proof of Theorem 9 (Route 8: inverse of log)

1	StrictConcaveOn $\mathbb{R}$ (Set.Ioi 0) Real.log	LogConc	Pos:18
2	StrictMonoOn Real.log (Set.Ioi 0)	LogMono	Pos:31
3	$\forall x : \mathbb{R},\ \texttt{Real.log}\ (\texttt{Real.exp}\ x) = x \wedge (\forall x : \mathbb{R},\ 0 < x \rightarrow \texttt{Real.exp}\ (\texttt{Real.log}\ x) = x)$	ExpLogEq	Pos:30
4	StrictConvexOn $\mathbb{R}$ (Real.log '' Set.Ioi 0) Real.exp	LogInv	Pos:19(3, 2, 1)
5	Real.log '' Set.Ioi 0 = Set.univ	surj	Eq(3) ∧ Pos:23
6	StrictConvexOn $\mathbb{R}$ Set.univ Real.exp	subst	Eq(4, 5)
7	ConvexOn $\mathbb{R}$ Set.univ Real.exp	weaken	Eq(6)
8	$\top$	close	→ I(1, . . . , 7)

Proof of Theorem 10 (Route 9: Lean pointer)

1	StrictConvexOn $\mathbb{R}$ Set.univ Real.exp per Lean's proof (Route 10)	Ref	Thm:11
2	ConvexOn $\mathbb{R}$ Set.univ Real.exp	StrictConvexOn.convexOn	Lean
3	$\top$	close	→ I(1, . . . , 2)

Proof of Theorem 11 (Route 10: Lean slope, detailed)

Left inequality. For $x < y$ in $\mathbb{R}$:

1	$x < y$ in $\mathbb{R},\ h := y - x > 0$	hypothesis	Hyp
2	$\forall t : \mathbb{R},\ t \neq 0 \rightarrow \texttt{Real.exp}\ t > 1 + t$	StrSupp	Pos:20
3	$\texttt{Real.exp}\ (-h) > 1 - h$	apply	Eq(2) ∧ 1
4	$1 - \texttt{Real.exp}\ (-h) < h$	rearrange	Eq(3)
5	$\texttt{Real.exp}\ y - \texttt{Real.exp}\ x = \texttt{Real.exp}\ y \cdot (1 - \texttt{Real.exp}\ (-h))$	mult	Pos:5(1)
6	$\texttt{Real.exp}\ y - \texttt{Real.exp}\ x < \texttt{Real.exp}\ y \cdot h = \texttt{Real.exp}\ y \cdot (y - x)$	combine	Eq(4, 5)
7	$(\texttt{Real.exp}\ y - \texttt{Real.exp}\ x)/(y - x) < \texttt{Real.exp}\ y$	divide	Eq(6)

Right inequality. For $y < z$ in $\mathbb{R}$:

8	$y < z$ in $\mathbb{R},\ k := z - y > 0$	hypothesis	Hyp
9	$\texttt{Real.exp}\ k > 1 + k$	apply	Eq(2) ∧ 8

10	$\texttt{Real.exp}\ k - 1 > k$	**rearrange**	**Eq(9)**
11	$\texttt{Real.exp}\ z - \texttt{Real.exp}\ y = \texttt{Real.exp}\ y \cdot (\texttt{Real.exp}\ k - 1)$	**mult**	**Pos:5(8)**
12	$\texttt{Real.exp}\ z - \texttt{Real.exp}\ y > \texttt{Real.exp}\ y \cdot k = \texttt{Real.exp}\ y \cdot (z - y)$	**combine**	**Eq(10, 11)**
13	$\texttt{Real.exp}\ y < (\texttt{Real.exp}\ z - \texttt{Real.exp}\ y)/(z - y)$	**divide**	**Eq(12)**

Combination.

14	$x < y < z \to (\texttt{Real.exp}\ y - \texttt{Real.exp}\ x)/(y - x) < \texttt{Real.exp}\ y < (\texttt{Real.exp}\ z - \texttt{Real.exp}\ y)/(z - y)$	**chain**	**Eq(7, 13)**
15	**adjacent slopes strictly increasing**	**slope**	**Def:6(14)**
16	**StrictConvexOn** $\mathbb{R}$ **Set.univ** $\texttt{Real.exp}$	**SlopeCrit**	**Pos:21(15)**
17	$\top$	**close**	$\to$ **I(1, . . . , 16)**

6.6.1. Dependency trees

Each row has the form $\texttt{L}n \mid \mathit{entry}\ \texttt{(strength)}$**:** n **is the depth from the root** $L0$**, and strengths are** `FOL`**,** `RCA0`**,** `WKL0`**,** `ZF`**.**

Route 1 (weighted AM–GM)

```
L0 | Conv_AMGM                    (RCA0)
L1 |   Pos:AMGM_R                 (RCA0)
L2 |     Pos:AMGM_Q               (RCA0)
L1 |   Pos:ExpMult                (RCA0)
L2 |     Pos:SeriesCauchy         (RCA0)
L2 |     Def:exp                  (RCA0)
L1 |   Pos:ExpPow                 (RCA0)
L2 |     Pos:ExpMult              (RCA0)
L2 |     Def:exp                  (RCA0)
L1 |   Def:exp                    (RCA0)
```

Route 2a (direct series on $\mathbb{R}_{\geq 0}$)

```
L0 | Conv_Ser                     (RCA0)
L1 |   Pos:Monomial               (RCA0)
L2 |     ax:conv-mul-mon          (ZF)
L2 |     ax:conv-order            (ZF)
L1 |   Pos:SeriesCauchy           (RCA0)
L1 |   Def:exp                    (RCA0)
```

Route 2b (translation to $\mathbb{R}$)

```
L0 | Conv_Trans                   (RCA0)
L1 |   Conv_Ser                   (RCA0)  -- see 2a
L1 |   Pos:Arch                   (RCA0)
L1 |   Pos:ExpMult                (RCA0)
L2 |     Pos:SeriesCauchy         (RCA0)
L2 |     Def:exp                  (RCA0)
L1 |   Pos:ExpPos                 (RCA0)
L2 |     Pos:SeriesCauchy         (RCA0)
L2 |     Def:exp                  (RCA0)
L1 |   Def:exp                    (RCA0)
```

Route 3 (midpoint convexity)

```
L0 | Conv_Mid                        (RCA0)
L1 |   Pos:Midpoint                  (RCA0)
L2 |     Pos:ExpMult                 (RCA0)
L2 |     ax:conv-mul-mon             (ZF)
L1 |   Pos:ExpCont                   (RCA0)
L2 |     Pos:SeriesCauchy            (RCA0)
L1 |   Pos:Mid2Conv                  (RCA0)
L2 |     Def:exp                     (RCA0)
L1 |   Pos:SeqCont                   (RCA0)
L1 |   Def:exp                       (RCA0)
```

Route 4 (second-derivative test)

```
L0 | Conv_2nd                        (WKL0)
L1 |   Pos:Conv2nd                   (WKL0)
L2 |     Pos:DerPosMono              (WKL0)
L3 |       Pos:conv-FTC              (WKL0)
L3 |       Pos:conv-int-strict       (WKL0)
L2 |     Pos:ConvDerStr              (WKL0)
L3 |       Pos:conv-FTC              (WKL0)
L3 |       Pos:conv-int-strict       (WKL0)
L1 |   Pos:ExpC2                     (RCA0)
L2 |     Pos:SeriesCauchy            (RCA0)
L2 |     Def:exp                     (RCA0)
L1 |   Pos:ExpDeriv                  (RCA0)
L2 |     Pos:SeriesCauchy            (RCA0)
L2 |     Def:exp                     (RCA0)
L1 |   Pos:ExpPos                    (RCA0)
L2 |     Pos:SeriesCauchy            (RCA0)
L2 |     Def:exp                     (RCA0)
L1 |   Def:exp                       (RCA0)
```

Route 5 (derivative strictly increasing)

```
L0 | Conv_Deriv                      (WKL0)
L1 |   Pos:ConvDerStr                (WKL0)
L2 |     Pos:conv-FTC                (WKL0)
L2 |     Pos:conv-int-strict         (WKL0)
L1 |   Pos:DerPosMono                (WKL0)
L2 |     Pos:conv-FTC                (WKL0)
L2 |     Pos:conv-int-strict         (WKL0)
L1 |   Pos:ExpC2                     (RCA0)  -- needed by DerPosMono
L2 |     Pos:SeriesCauchy            (RCA0)
L2 |     Def:exp                     (RCA0)
L1 |   Def:conv-Ck                   (RCA0)  -- C2 -> C1, used with Pos:ExpC2
L1 |   Pos:ExpDeriv                  (RCA0)
L2 |     Pos:SeriesCauchy            (RCA0)
L2 |     Def:exp                     (RCA0)
L1 |   Pos:ExpPos                    (RCA0)
L2 |     Pos:SeriesCauchy            (RCA0)
L2 |     Def:exp                     (RCA0)
L1 |   Def:exp                       (RCA0)
```

Route 6 (supporting-line inequality)

```
L0 | Conv_Supp                    (RCA0)
L1 |   Pos:EZ                     (RCA0)
L2 |     Pos:SeriesCauchy         (RCA0)
L2 |     Def:exp                  (RCA0)
L1 |   Pos:Support                (RCA0)
L1 |   Pos:ExpMult                (RCA0)
L2 |     Pos:SeriesCauchy         (RCA0)
L1 |   Pos:ExpPos                 (RCA0)  -- for e^y > 0
L2 |     Pos:SeriesCauchy         (RCA0)
L1 |   Def:exp                    (RCA0)
```

Route 7 (limit definition and Bernoulli)

```
L0 | Conv_Bern                    (RCA0)
L1 |   Pos:LimitExp               (RCA0)
L2 |     Pos:SeriesCauchy         (RCA0)
L2 |     Def:exp                  (RCA0)
L1 |   Pos:Bernoulli              (RCA0)
L1 |   Pos:Support                (RCA0)
L1 |   Pos:ExpMult                (RCA0)
L2 |     Pos:SeriesCauchy         (RCA0)
L1 |   Pos:ExpPos                 (RCA0)  -- for e^y > 0
L2 |     Pos:SeriesCauchy         (RCA0)
L1 |   Def:exp                    (RCA0)
```

Route 8 (inverse of the concave logarithm)

```
L0 | Conv_Log                     (RCA0)
L1 |   Pos:LogConc                (RCA0)  -- via weighted AM-GM
L2 |     Pos:AMGM_R               (RCA0)
L1 |   Pos:ExpLogEq               (RCA0)  -- inverse characterisation of log and exp
L2 |     Def:exp                  (RCA0)
L2 |     Pos:StrSupp              (RCA0)  -- injectivity of exp, for log(exp y) = y
L2 |     Pos:Arch                 (RCA0)  -- for surjectivity of exp on (0, infty)
L1 |   Pos:LogMono               (RCA0)  -- exp^{-1} strictly increasing
L2 |     Pos:ExpLogEq            (RCA0)  -- as above
L2 |     Pos:StrSupp              (RCA0)  -- as above
L2 |     Pos:ExpMult              (RCA0)
L2 |     Pos:ExpPos               (RCA0)
L1 |   Pos:LogInv                (RCA0)
L2 |     Pos:LogConc              (RCA0)  -- as above
L2 |     Pos:LogMono              (RCA0)  -- as above
L2 |     Pos:ExpLogEq             (RCA0)  -- as above
```

Route 9 (Lean's pointer)

```
L0 | Conv_Lean                    (RCA0)
L1 |   Conv_LeanDet               (RCA0)  -- see Route 10
```

Route 10 (Lean slope criterion, detailed)

```
L0 | Conv_LeanDet                 (RCA0)
L1 |   Pos:StrictSupport          (RCA0)
```

```
L2 |     Pos:SeriesCauchy              (RCA0)
L2 |     Def:exp                       (RCA0)
L1 |   Pos:SlopeCrit                   (RCA0)
L1 |   Pos:ExpMult                     (RCA0)
L2 |     Pos:SeriesCauchy              (RCA0)
L1 |   Def:Conv                        (RCA0)
```

6.6.2. Depth and strength

Route(s)	Max depth	Depth chain	Strength
1,2a,2b,3,6,7,8,9,10	≤ 2	—	RCA0
4	3	Conv2nd $\to$ {DerPosMono, ConvDerStr} $\to$ {FTC, IntStr}	WKL0
5	2	{ConvDerStr, DerPosMono} $\to$ {FTC, IntStr}	WKL0

Entry	Table tag	Bishop–Cauchy	In tree(s)
ax:conv-mul-mon	ZF	RCA_0	2a, 3
ax:conv-order	ZF	RCA_0	2a, 3
Pos:FTC	WKL_0	WKL_0	4, 5
Pos:IntStr	WKL_0	WKL_0	4, 5
Pos:MVT	WKL_0	—	none

6.6.3. Strength map

#	Route	Lower	Upper	Choice
1	Weighted AM–GM	RCA_0	RCA_0	none
2a	Direct series on $\mathbb{R}_{\geq 0}$	RCA_0	RCA_0	none
2b	Translation to $\mathbb{R}$	RCA_0	RCA_0	none
3	Midpoint convexity	RCA_0	RCA_0	none
4	Second-derivative test	open	WKL_0	none
5	Derivative strictly increasing	open	WKL_0	none
6	Supporting line via series	RCA_0	RCA_0	none
7	Limit definition + Bernoulli	RCA_0	RCA_0	none
8	Inverse of concave logarithm	RCA_0	RCA_0	none
9	Lean Mathlib (pointer)	RCA_0	RCA_0	none
10	Lean Mathlib slope (detailed)	RCA_0	RCA_0	none

Claim	Justification
Routes 4–5 classify inference patterns	thmi : conv — 2nd, thmi : conv — deriv RCA0 via implication-introduction: antecedent in {Pos : ExpDeriv, Pos : ExpPos}; consequent in {thm : series — pos, thm : translation, thm : midpoint — convex}. RCA_0 reformulation of patterns 4–5 for exp.
exp RCA_0 provable via Routes 1,2,3,6,7,8,9,10.	No Ax : conv — complete, no ACA_0, no $\mathsf{DC}(\mathbb{R})$, no choice.
Routes 4–5 have WKL0 upperbound.	Carried by Pos : FTC, Pos : Int Str [5],IV.2.3; inherited by Conv2nd, ConvDerStr, DerPosMono. MVT, PDerMon, invoked by no route.

The deepest chains are Routes 4 and Route 5, at depth 3, through

$$\mathsf{Conv2nd}(\text{or } \mathsf{ConvDerStr}) \to \mathsf{DerPosMono} \to \mathsf{conv-FTC/conv-int-strict}.$$

These chains have strength WKL_0, since the two FTC tools at the bottom are WKL_0 over RCA_0 ([5, II.7, IV.2.3]).

- Routes 1, 2a, 2b, 3, 6, 7, 8, 9, 10 reach depth 2 at most, and their effective strength is RCA_0. The field axioms **ax:conv-mul-mon** and **ax:conv-order** are tagged **ZF** in the axiom table only to record that they are realized by **ZF** alone (no choice) under the Bishop–Cauchy coding of §6.1 they are theorems of RCA_0. The WKL_0 entry **MVT** appears in no tree.
- Routes 4 and 5 use the FTC-based tools **conv-FTC** and **conv-int-strict** directly, which are themselves WKL_0 (the same strength as **MVT**, but a strictly more elementary formulation).

For Routes 1, 2, 3, 6, 7, 8, 9, 10 the lower and upper bounds coincide: these routes have explicit RCA_0 proofs in §6.6 or are one-step consequences of RCA_0-postulates (**Support** for Route 6, **Bernoulli** for Route 7, **AMGM_R** + **LogInv** for Route 8). Routes 4 and 5 have WKL_0 upper bounds because

$$\mathsf{conv-FTC}, \mathsf{conv-int-strict}, \mathsf{DerPosMono}, \mathsf{ConvDerStr}, \mathsf{Conv2nd}$$

are all WKL_0 over RCA_0. The corresponding lower bounds (i.e. whether Routes 4 and 5 are RCA_0-provable) are not claimed here and would require a separate non-provability argument.
The theorem (convexity of exp) is RCA_0-provable via any of Routes 1, 2, 3, 6, 7, 8, 9, 10.
None of the routes uses the Cut axiom; none uses a choice principle.

6.6.4. Comparison

We distinguish four classes:

(1) Constructive in Bishop's sense (Routes 1–3).
(2) Classical differential (Routes 4–6),
(3) Further classical or analytic (Routes 7–8),
(4) Formal Lean (Routes 9–10).

The reverse-mathematical classification refines this taxonomy:
Routes 1, 2, 3, 6, 7, 8, 9, 10 are RCA_0-provable (the corresponding statements are theorems of RCA_0; whether they are equivalent to, or weaker than, RCA_0 is not at issue),
while Routes 4 and 5 sit at WKL_0 (upper bound; lower bound not claimed).
The reason for the gap is the classical differential machinery:

- **conv-FTC** (the FTC for continuously differentiable functions on $[a, b]$) is WKL_0 over RCA_0 ([5, IV.2.3]).

- **conv-int-strict** (strict monotonicity of the Riemann integral) inherits the same strength, because it is derived from the FTC.
- **conv-2nd**, **conv-deriv-mono**, **conv-deriv-strict** are each WKL_0 consequences of the above two.
- **DerMon** (monotone derivative $\Rightarrow$ convex) is WKL_0 as well, though it is not invoked by any route.

The classical MVT is WKL_0 over RCA_0 ([5, II.7]), but is not used by any route: the FTC-based derivations sidestep it.

The Lean routes, provided the cited Lean proof avoids `classical.choice` and other non-constructive principles, can be interpreted in RCA_0; their classification therefore coincides with the constructive routes they formalise. We do not claim this interpretability for every proof in Mathlib, only for the specific proof cited here.

Consequently:

The convexity of `Real.exp` is RCA_0-provable; some of its classical proof routes (Routes 4 and 5) are WKL_0-provable (upper bound).

The theorem is thus RCA_0.

The proof routes that use classical differential machinery have upper bound WKL_0.

No route needs ACA_0, $\mathsf{DC}(\mathbb{R})$, or any choice principle.

The Dedekind Cut axiom is superfluous because the series carries an explicit modulus of convergence.

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